STEP Questions & Commentary

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1 Introduction

This document will exhaustively identify every STEP question I have covered since I started preparing for STEP in November 2026, and what I have learned from each one. For each question, I have uniquely identified the question, where I got it from as well as my thoughts and commentary on it.

I decided to sit STEP in my gap year to target universities such as Warwick, UCL and potentially Imperial College London. I have been doing these questions alongside various problems from BMO1.

2 Siklos' Problem 7: Chain of Equations (1997 STEP II)

Having had done the first few questions of Siklos' Advanced Problems in Mathematics book an absurd number of times, I thought I would start my preparation with a question I hadn't seen before. As Siklos' own commentary suggests, this question leaves you to your own devices, but is extremely easy with some pattern recognition. The emerging pattern:

$$x_{n+1} = \frac{2^{k+1} - 2}{2^{k+1} - 1}$$

is near identical to one of the PAT Question 1s. Having sat the PAT in October 2025, recognising the underlying structure was the simple part. The actual induction proof was slightly interesting given the manipulation of algebraic fractions within the denominator but was overall quite straightforward.

3 Siklos' Problem 4: Divisibility (1999 STEP I)

This question had some simple ideas and some more complex ideas which I believe would make it really good for any Computer Science aspirants as well as anyone sitting the Further Pure 2 exam.